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Dr. Carsten Lemmen
Institute of Coastal System Analysis and Modeling
HEREON

$HEREON Address - if available$

Dear Dr. Lemmen,
I am writing to express my strong interest in the Postdoctoral Researcher position in hydrodynamic-biogeochemical modeling at HEREON. The AGRIO project's focus on greenhouse gas emissions in the Elbe-North Sea system aligns perfectly with my research interests and skills.

After my M.Sc. in Physics and following the completion of my Ph.D. in astrochemistry, I transitioned to ecosystem modeling and joined the group of Prof. Dr. Inga Hense as a postdoctoral researcher. During this time, I gained extensive practical experience in 0D IBM (individual based model), compartment-based 1D hydrodynamic-biogeochemical modeling, specifically working with the GOTM hydrodynamic model coupled with the FABM biogeochemical framework. My recent work involves implementing trait diffusion of temperature optimum (Beckmann et al. (2019)) developing and implementing it in a Cyanobacteria Life Cycle (CLC) model which is embedded in the ERGOM model (Neumann 2000). This, combined with experimental temperature response data of Dolichospermum over many years from resurrection experiments, enables me to perform hindcast and projection studies into future warming climatic scenarios. This work is still in progress, both experimentally and in terms of model studies. This experience has provided me with a strong foundation in the modeling techniques that are directly relevant to the AGRIO project and the research conducted at HEREON in your group. Also, being a Fortran enthusiast and realizing the immense power of Python (still a beginner) in analyzing NetCDF data (notwithstanding FERRET) gives me extra impetus to apply for this position.

I am very comfortable working in Linux environments, where I work daily, and have experience in developing and running simulations, and visualizing large datasets. My PhD work included analyzing terabytes of collision experiments data. I am confident that my skills in numerical modeling, combined with my eagerness to learn and adapt, make me a strong candidate for this position.

I am particularly drawn to the toolset GOTM-FABM, in fact, I am very engaged with it these days, understanding how powerful the FABM framework is in coupling to a hydrodynamic model such as GOTM. It is a wonderful opportunity to extend my experience, develop new models or modify existing ones, and to contribute to research that addresses critical environmental challenges. I am eager to apply my skills to investigate the impacts of anthropogenic stressors and climate change on GHG dynamics in the Elbe-North Sea system.

I have attached my CV for your review and welcome the opportunity to discuss my qualifications further in an interview.

Sincerely,
Your Name